

Operational Research

Introduction to Simulation Answers

The following exercises will require the use of Excel. An example question and excel spreadsheet is given on Course Resources.

Exercises

- 1) By Hand Complete the following table

Customer Arrival Time	Service Starts	Service Duration	Service Ends	Customer Waiting Time	Service Waiting Time
4		3			
6		3			
12		4			
14		2			
17		3			

Handwritten solution for Exercise 1:

Customer arrival time	Service Starts	Service Duration	Service ends	Customer waiting time	Service waiting time
4	4	3	7	0	4
6	7	3	10	1	0
12	12	4	16	0	2
14	16	2	18	2	0
17	18	3	21	1	0

- 2) Data has been collected at a service centre to enable the analysis and redesign of the service system.

Customers arrivals are described by the distribution

Inter-arrival Times	5	10	15	20	25	30
Frequency percentage	10	20	40	20	7	3

Service times are given by

Service Duration	10	12	14	16	20	30
Frequency percentage	5	20	45	20	9	1

Simulate the arrival of 30 customers, assuming that only the quoted times could occur, what does your simulation indicate about this system? What could be done to improve the system? *

Customer	Inter-arrival Time (IAT)	Arrival Time (AT)	Service Starts (SS)	Service Duration (SD)	Service Ends (SE)	Customer Waiting Time	Service Waiting Time
1	Generate using random number	IAT(1)	AT(1)	Generate using random number	SS(1) + SD(1)	SS(1) – AT(1)	SS(1)
2	Generate using random number	AT(1) + IAT(2)	Max{SE(1), AT(2)}	Generate using random number	SS(2) + SD(2)	SS(2) – AT(2)	SS(2) - SE(1)

3) Jobs arrive every 15 minutes at a workshop comprising a single production stage.

Job arrivals are described by the distribution

Number of jobs arriving	5	7	11	12
Frequency percentage	10	20	40	30

Service capacities (i.e. the number of jobs processed per 15 minutes) are given by

Jobs Capacity (Service Capacity)	6	8	10	15
Frequency percentage	15	40	35	10

Simulate the production process over a period of 4 hours, what does your simulation indicate about this system? What could be done to improve the system? **

Time	No. Jobs Arriving (JA)	Jobs in Queue (JQ)	Jobs Capacity (JK)	Jobs Completed (JC)	Total Jobs Completed (TJC)	Total Jobs Left (TJL)	Unused Capacity (UC)
1 (0-15)	Generate using random number	JA(1)	Generate using random number	Min{JQ(1),JK(1)}	JC(1)	JQ(1)-JC(1)	JK(1)-JC(1)
2 (15-30)	Generate using random number	JA(2)+ TJL(1)	Generate using random number	Min{JQ(2),JK(2)}	JC(1)+JC(2)	JQ(2)-JC(2)	JK(2)-JC(2)